



RAPHAEL PENZ

📍 St. Anton, Tirol, Austria | ✉ raphpenz@gmail.com | 📞 +43 676 589 4674 | 🌐 theh0bbit.github.io/

SUMMARY

Games Engineering graduate from the Technical University of Munich with experience developing and releasing a large-scale solo RPG/Visual Novel project. Strong background in gameplay systems, software architecture, Python, and C++, complemented by experience in high-performance simulation development through academic research. Interested in gameplay programming, systems design, and software development roles.

PROJECT EXPERIENCE

Harmony of the Stars (Game Project)

September 2024 - Present

- **Solo Developer** | theh0bbit.github.io/
 - Designed and developed a solo RPG / Visual Novel hybrid using Python and Ren'Py.
 - Implemented custom gameplay frameworks including combat, inventory, quests, progression, relationships, scheduling, and world navigation.
 - Designed and implemented a modular software architecture supporting interconnected gameplay systems and long-term maintainability.
 - Developed over 23,000 lines of gameplay code and 47,000 lines of total project code.
 - Released the project publicly, reaching over 2,500 cumulative downloads.
 - Designed and implemented a save-compatible architecture, enabling future game updates without breaking existing save files.
 - Solely responsible for programming, gameplay systems, software architecture, worldbuilding, and release management.

Implementation and Evaluation of Additional Particle Simulation Types with AutoPas (Bachelor Thesis)

October 2021

- **Technical University of Munich**
 - Implemented a new Discrete Element Method (DEM) particle type in the AutoPas C++ particle simulation framework.
 - Developed pairwise force calculation functors supporting AoS and SoA memory layouts with SIMD optimization.
 - Extended simulation infrastructure and integrated DEM simulations into the existing molecular dynamics benchmark suite.
 - Evaluated scalability and performance on simulations containing up to 1,000,000 particles.
 - Compared physical behavior and runtime characteristics against existing molecular dynamics implementations.

EDUCATION

**M.Sc. Computer
Science: Games
Engineering**

- **Technical University of Munich (TUM)**
Munich • Oct 2024 - Sept 2025
• Degree not completed

**B.Sc. Computer
Science: Games
Engineering**

- **Technical University of Munich (TUM)**
Munich • Oct 2017 – Sept 2024

WORK EXPERIENCE

**Hotel Assistant (Part-
Time)**

August 2021 - Present

- **Luxury Boutique Hotel Himmlhof** | St. Anton am Arlberg, Tirol
 - Supported day-to-day hotel operations in a part-time capacity.
 - Provided IT support, including workstation setup, software troubleshooting, printer configuration, website updates, and general technical maintenance.
 - Assisted with guest services, reception operations, housekeeping, and general operational tasks.
 - Adapted to varying responsibilities depending on seasonal and operational requirements.

**Civilian Service /
Emergency Medical
Technician (EMT)**

September 2016 - June 2017

- **Austrian Red Cross** | Landeck, Tirol
 - Served as a paramedic
 - Supported dispatch operations, including call handling, scheduling patient transport services, and coordinating drivers and vehicles.

SKILLS

Programming Languages

C++ • Python • C# • Java

Tools & Engines

Git • Ren'Py (Advanced) • Unity (Intermediate) • Unreal Engine (Basic Familiarity) • GIMP • Photoshop • DaVinci Resolve • Audacity

Software Development

Object-Oriented Design • Gameplay Systems • Data-Driven Development • Save & Update-Compatible Architecture

Languages

German (Native) • English (C2)